PROPOSAL ON
CASE STUDY OF CURRENT STATUS OF SETI RIVER SUSPENDED BRIDGE

A Project Proposal submitted in partial fulfilment of the requirements of Pokhara University for the degree of Bachelor of Engineering.

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BONAFIDE CERTIFICATE

It is certified that this project I titled "**CASE STUDY OF SETI RIVER SUSPENDED BRIDGE**" is the bonafide work of – **Mr. Dipesh Bahadur Hamal, Mr. Hemanta Aryal, Mr. Krish Prasad Jaiswal, Mr. Prasanna Pahari, Mr. Saurav Giri, Mr. Sujan Lamichhane and Mr. Suyog Poudel** who carried out the research/project work under my supervision. It is further certified that to the best of my knowledge the work reported herein doesn't form part of any other thesis or dissertation or project on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

As prescribed by Pokhara University syllabus “Rural Project I” in sixth semester serves as an effective to get acquainted with practical problems that one has to face as an application of various theoretical knowledge.

Our project I is on the case study of the components of Seti River Suspended Bridge, Pokhara that links Pokhara metropolis-30 and Pokhara metropolis -33. We are going to study the components of the bridge and their dimensions. Then we are going to follow different methodology for the completion of our project. Different statements of problems with remedy measures are going to be discussed and we will conclude to look after this bridge to respective authority.

Keywords: **suspended bridge, components, case study**

TABLE OF CONTENTS

ACKNOWLEDGEMENT……I
ABSTRACT……II
TABLE OF CONTENTS……III
LIST OF FIGURES……IV
LIST OF TABLES ……V
ABBREVIATIONS……VI

TITLE……1

1. Introduction 1
1.1 Background……1
1.2 Components of Suspended trial bridge……1
1.3 Location……4
1.4 Load transfer Mechanism……4
2. Statement of problems.....5
3. Necessity (Significance).....5
4. Objectives 5
4.1 Specific Objective 5
4.2 General Objective 5
5. Literature Review ……5
6. Scope..... 6
7. Methodology 6
7.1 Group Formation 6
7.2 Group Discussion 6
7.3 SiteVisit..... 6
7.4 Data Collection 6
7.5 Report Writing …….....6
8. Expected Result..... 7
9. Significance of Expected result.....8
10. Cost Estimation.....8
11. Work Plan……8
12. Conclusion…….....8
13. References.....9

LIST OF FIGURES

Figure No.	Title	Page no.
1.	Suspended trial bridge	3
2.	Section selected for case study of Seti river bridge	4
3.	Flowchart of methodology	7

LIST OF TABLES

Table No.	Title	Page no.
1.	Different Components of bridge	2
2.	Cost Estimation	8
3.	Work Plan	8

ABBREVIATIONS

Symbols and Abbreviations		Meaning
BCRE	-	Bachelor of Civil and Rural Engineering
SOE	-	School of Engineering
HMG	-	His Majesty's Government
SBD	-	Suspension Bridge Division
BBLL	-	Bridge Building at Local Level
PU	-	Pokhara University

1. Introduction

Suspended trail bridge is a type of bridge in which the deck is hung below the hand rail cable on the main cables in which wind guy cables and wind tie cables are used to provide lateral stability. The introduction of suspended trail bridges in Nepal marks a significant evolution in infrastructure development aimed at enhancing connectivity and access in mountainous regions.

1.1. Background

Historically, prior to 1964/65, development operations in Nepal were limited, particularly during the Rana era, with only a few bridges utilizing imported components from Scotland. These early bridges, situated on major routes, had minimal impact on local communities (HMG and HELVETAS, 1989).

In response to the growing need for reliable pathways across Nepal's rugged terrain, the government established the Suspension Bridge Division (SBD) in 1964. This initiative aimed to address local transportation challenges more effectively (HMG and HELVETAS, 1989). Despite initial efforts, the SBD struggled to meet the widespread demand for trail bridges, resulting in the construction of bridges that often fell short of functional expectations.

Recognizing these limitations, a strategic shift occurred in 1990, emphasizing community-driven approaches to trail bridge construction. This pivotal moment led to the inception of the Bridge Building at the Local Level (BBLL) program, supported by the Swiss government through HELVETAS Nepal. The BBLL program aimed to empower local communities to actively participate in the planning, construction, and maintenance of trail bridges, thereby ensuring bridges were both functional and sustainable (HMG and HELVETAS, 1989).

1.2 Components of suspended trail bridge:

- Foundation
- Abutment wall
- Steel deck
- Hand rail cable
- Main Cable
- Suspender
- Wire mesh
- Wind guy cable
- Wind tie cable

1.2.1 Foundation: The foundation of a suspended trail bridge is a critical component of the structure, it provides the support and stability necessary to withstand the loads and stresses placed on the bridge. The foundation is the part on which the link wall rest.

1.2.2. Abutment wall: In a bridge, the abutment walls are adjacent to the foundation and act as retaining walls. They are generally constructed of same material as of foundation. The abutment walls can either be attached to the foundation or be independent of it. It is a support wall over which hand rail cables are passed and are connected to foundation.

1.2.3. Steel deck: Steel deck, also known as metal deck is a sheet of high-performance galvanized steel laid over the main cables. It supports the traffic and the live load on the bridge.

1.2.4. Hand rail cable: Hand rail cable is generally made up of many steel wires spun together at construction sites which is tied on the both ends of the link walls to the foundation of the bridge. It is typically placed (generally 2) at hand height or slightly higher for people to hold for support to safely ascend or descend stairs ramps or move across the platform. The main purpose of hand rail cable is to offer a firm grip and aid in safe and careful movements.

1.2.5. Main cables: Main cables are similar to the hand rail cable in appearance as well as composition but it bears more load than that of hand rail cable. In suspended bridge, main cables (normally 6) hang between the two foundation and are anchored at each end to the foundation. Main cables which are laid below the steel deck are under tension from their own weight.

1.2.6. Suspender: Suspender is a small thin rod made up of steel which connects hand rail cable and main cable to maintain constant shape of the bridge.

1.2.7. Wire Mesh: It is a mesh like structure of steel wires which is tied between hand rail cables and main cable along with linkers. The main purpose of wire mesh is safety.

1.2.8. Wind tie cable: Wind tie cables are provided to connect main cables to the wind guy cable with the main purpose of providing lateral stability to the bridge. They are made up of steel wires spun together as a bundle.

1.2.9. Wind guy cable: Wind guy cables are the cables attached to the wind tie cables in the parabolic form providing lateral stability to the bridge. They are anchored on the guy foundation (generally 4 foundation) on the respective ends.

Table 1: Different Components of bridge

S.N.	Components	Description
1.	Foundation	At the ends of bridges span which provides support and stability
2.	Abutment wall	Lies on the foundation over which hand rail cable passes
3.	Steel deck	Placed on the main cables for transportation purpose
4.	Hand rail cable	Over the main cables connected through link wall to foundation

5.	Main cable	Below the steel deck to bear load
6.	Suspender	Vertical connectors of hand rail cable and main cable
7.	Wire mesh	Between linkers for safety
8.	Wind tie cable	Connects main cable to wind guy cable providing lateral stability
9.	Wind guy cable	Parabolic cable connected to tie cable between guy foundation on both ends

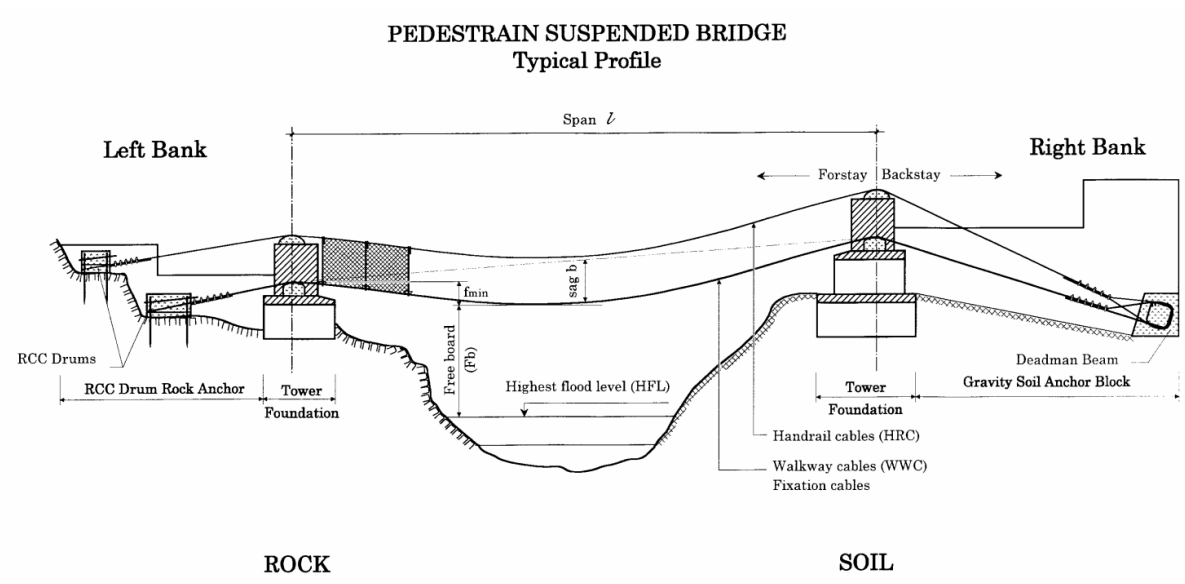

Fig 1: Suspended trial bridge

Source: D Type Handbook.pdf (nepaltrailbridges.org.np)

1.3. **Location**: Project is going to be held on Seti bridge (a suspended trial bridge which carries deck on top of main cables) that links Pokhara-30 and Pokhara-33.

Fig 2: Section selected for case study of Seti river bridge

1.4 Load transfer mechanism:

In a suspended bridge, the load transfer mechanism works as follows:

- The weight of the bridge deck is transferred to the cables that are suspended from the foundation supports.
- The cables are anchored at both ends of the bridge, and the weight of the deck is transferred to the foundation through the main cables.
- The foundation support the weight of the bridge deck by transferring the load to the ground.
- The shape of the cables is critical to the load transfer mechanism. The cables are designed to distribute the load evenly across the entire length of the bridge deck, and to minimize the stresses on the cables themselves.
- The deck of the bridge is typically made of steel and is designed to resist the loads imposed on it by traffic, wind, and other factors.
- The cables are made of high-strength steel or other materials, and are designed to resist the tension forces created by the weight of the bridge deck.
- The foundation are typically made of concrete and are designed to resist the compressive forces created by the weight of the bridge deck.

Overall, the load transfer mechanism of a suspended bridge relies on the balance of forces between the weight of the bridge deck and the tension in the cables that support it. Proper design and construction are critical to ensure that the bridge is safe and stable under all conditions.

2.Statement of Problems

There are various problems on different components of bridge such as corrosion, problem caused by fatigue load and many more. Such problems are seen due to atmospheric conditions, lack of maintenance and inspection.

3.Necessity (Significance)

The bridge of Seti river suspended trial Bridge links Pokhara metropolis-30 and Pokhara metropolis-33 is helping people to connect with each other from different areas and for different purposes.

- Mainly benefitted to the local people of Pokhara and connected areas.
- Provide safe passage.

4. Objective

4.1 Main objective

- To study the current status of bridge.

4.2 Specific objectives

- To study components of bridge in detail.
- To identify the defects present in the bridge.

5. Literature Review

For centuries, Nepalese communities have constructed simple and inexpensive bridges using locally sourced materials such as timber, bamboo, or steel (HMG and HELVETAS, 1989). However, with the advent of modern development, organizations have introduced more sophisticated suspension bridges where local resources seemed inadequate (HMG and HELVETAS, 1989). Over the last four decades, approximately five hundred of these advanced and costly bridges have been erected, significantly enhancing connectivity along major transportation routes in the country (HMG and HELVETAS, 1989).The development of the Short-Span Trail Bridge Standard began in 1989, initiated jointly by the Government of Nepal (HMG) and a Swiss development organization (HELVETAS). This initiative aimed to introduce a new approach to building community trail bridges in Nepal's hill areas, ensuring they are resilient to extreme weather and seismic activity (HMG and HELVETAS, 1989). Typically, the construction of these trail bridges is carried out collaboratively by the government, non-governmental organizations (NGOs), and local communities (HMG and HELVETAS, 1989). Their primary goal is to provide safe access to essential services such as healthcare, education, and markets, thereby promoting economic development in remote regions (HMG and HELVETAS, 1989).

6. Scope

This study is about the analysis of bridge and its components. In this study the components of bridge and defects will be addressed by visiting the site.

7. Methodology

In order to make our project effective and successful, the following methods will be applied.

7.1 Group Formation

At first, members for the group were chosen by discussing with friends and teachers and group was formed.

7.2 Topic selection and group discussion

After the formation of group a specific and suitable topic will be chosen. Then, we will discuss about the main theme of our project among the group members. Every member's views will be included and will make a plan on how to go on with the project.

7.3 Site Visit

After group discussion, we will visit the site for understanding its general features. It's a part of the quality evaluation process which will create an interactive learning environment for us to understand components of suspended trail bridge.

7.4 Data Collection

Primary as well as secondary data will be collected for the project studies but our main focus will be on primary data. Primary data will be collected from site visit. Secondary data collection will be carried out with the help of related past projects reports, neighboring residential area, published papers and journals and other relevant sources. Then, the data collected will be proceed and arranged in systematic ways in tabular form.

7.5 Report Writing

All the data and information will be managed in a sequential order. The findings of the project and final conclusions will be produced. Finally, recommendations will be provided.

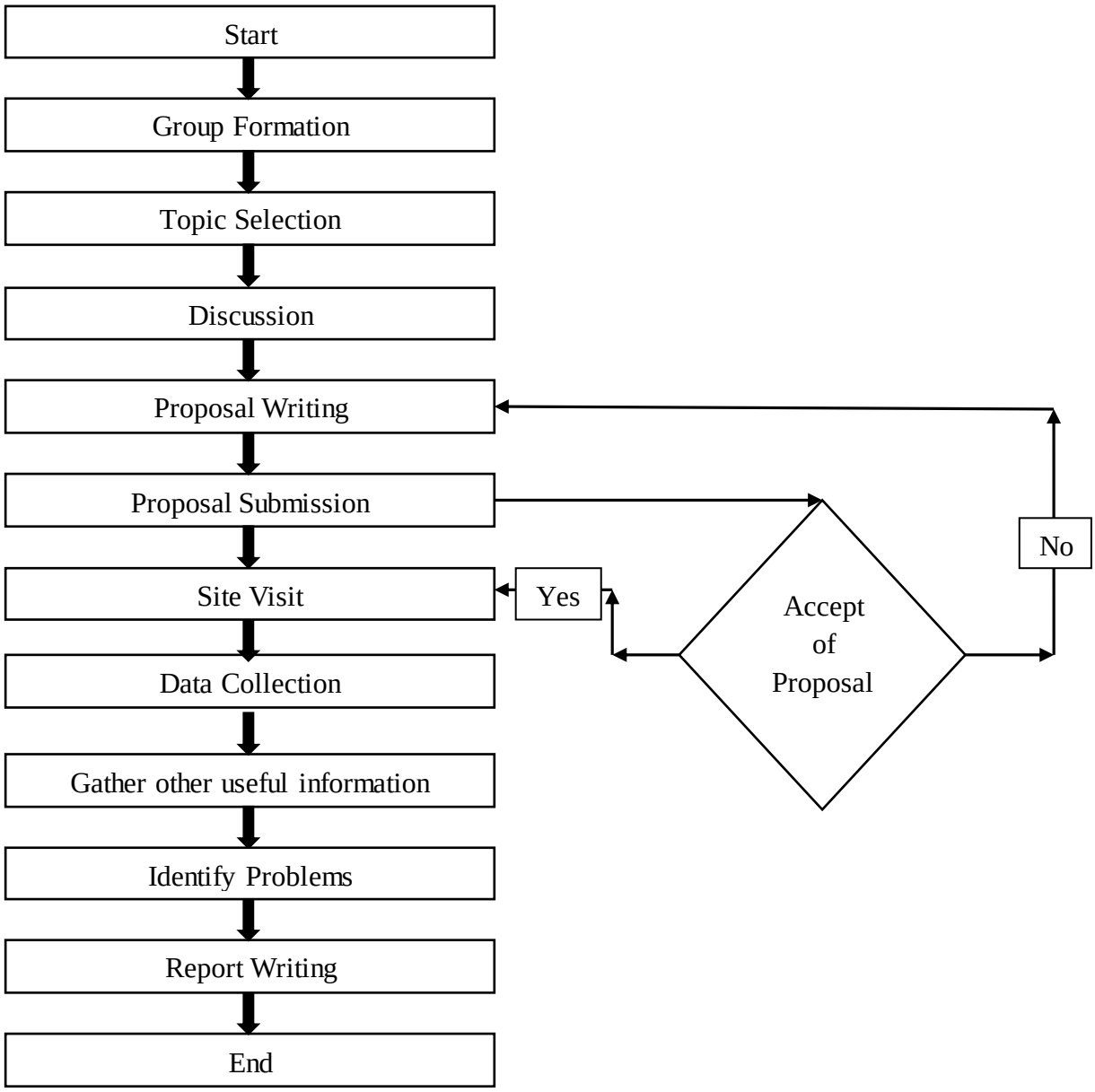

Fig 3: Flowchart of methodology

8. Expected Result

It is expected that we will be able to learn about different components of bridges. The methodology developed will develop a framework for the better collection and utilization of collected data and reports.

9. Significance of Expected Result

We will have a deeper understanding towards the importance of components of bridge. The findings of the project will help us to explore more ideas. This case study will also be the basic for our upcoming projects as it will help to gain required information.

10. Cost Estimation

Table 2: Cost Estimation

S.N	Contents	Costs
1	Transportation	800
2	Food	2500
3	Printing	1000
4	Miscellaneous	1000
Total		**5300**

11. Work Plan

Table 3: Work Plan

S.N	Activities	Duration (Days)															
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
1	Group Formation																
2	Topic Selection And Discussion																
3	Proposal Writing																
4	Proposal Approval																
5	Site Visit																
6	Data Collection																
7	Report Writing																
8	Report Submission																
9	Project Defence																

12. Conclusion

The following conclusions will be drawn:

• The components used for the construction of bridge will be clearly understood.

• The load transfer mechanism of suspended trail bridge will be known.

• A brief information about historical development of suspended trail in Nepal will be identified.

• The current status of bridge will be known.

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